

PIUM / SPARC κ-Screening and Baseline Scaling Analysis

First-pass selection of galaxies suitable for future morphological-coherence (κ) measurement using the accepted SPARC outer-slope sample

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Prepared from: pium_sparc_accepted_galaxies.csv + SPARC Table1 metadata
Image-retrieval coordinates were supplied separately for Spitzer SHA target lookup

1. Aim

This one evaluates whether the currently accepted SPARC outer-slope sample can support a future PIUM test based on the relation

$$s = a - b \log_{10}(V_{\text{flat}}) - c\kappa,$$

where κ is intended as a morphological-coherence parameter measured from 3.6 μm images. Because κ has not yet been measured, the present report performs the two necessary preparatory steps: (i) a geometry/size screening of the 79 accepted galaxies to identify systems suitable for a first-pass κ measurement, and (ii) baseline regressions without κ, in order to determine whether a later κ-term would be adding value to an already nontrivial scaling law.

2. Data sources and screening variables

The starting sample is the accepted 79-galaxy SPARC subsample used in the prior outer-slope recalculation. For each system the working table includes the outer slope, Vflat, inclination, distance, and Rdisk. The galaxy-by-galaxy global parameters are consistent with SPARC Table1 metadata. The user-supplied NED/SHA target resolutions provide coordinates for later image retrieval, but the numerical screening below uses only the SPARC structural parameters.

A first-pass κ measurement requires two practical conditions. First, the projected disk cannot be too edge-on or too face-on, otherwise an azimuthal Fourier decomposition becomes unstable or physically ambiguous. Second, the disk must be large enough on the sky for a meaningful low-order mode decomposition. The following operational cuts were adopted:

- inclination window: $30^\circ \leq \text{Inc} \leq 75^\circ$;
- apparent disk scale threshold: $\text{Rdisk,app} \geq 30 \text{ arcsec}$, where $\text{Rdisk,app} = \text{Rdisk(kpc)} \times 206265 / [1000 \times \text{D(Mpc)}]$.

These cuts are not claimed to be unique. They are intentionally conservative and are meant only to identify a reliable pilot sample for future κ extraction from Spitzer/IRAC 3.6 μm images.

3. Screening outcome

Applying the screening rules to the 79 accepted galaxies produces 28 systems that are geometrically suitable for a first-pass κ measurement. The remaining galaxies are excluded primarily because they are either too small on the sky for robust Fourier analysis or because their viewing geometry is too close to edge-on or face-on.

Screening category	Count
Excluded: too small on sky	28
Eligible for first-pass κ measurement	28
Excluded: edge-on geometry	13
Excluded: edge-on and too small	8
Excluded: face-on and too small	1
Excluded: face-on geometry	1

Two of the 28 screened-in galaxies have Vflat = 0 in the accepted table and therefore cannot enter the log10(Vflat) regressions. Accordingly, the baseline fits below use 26 κ-eligible galaxies.

4. Baseline regressions before κ

Four baseline models were fitted by ordinary least squares using outer_slope_ols as the dependent variable. The purpose is to measure how much of the slope structure is already explained by Vflat alone and whether a simple structural proxy such as Rdisk adds appreciable explanatory power before κ is introduced.

Sample	Model	n	Best-fit relation	R²
Full accepted sample	$s = a - b \log_{10}(V_{\text{flat}})$	77	$s = 1.272 - 0.571 \log_{10}(V_{\text{flat}})$	0.544
Full accepted sample	$s = a - b \log_{10}(V_{\text{flat}}) + d \log_{10}(\text{Rdisk})$	77	$s = 1.411 - 0.648 \log_{10}(V_{\text{flat}}) + 0.060 \log_{10}(\text{Rdisk})$	0.550
κ-eligible subset	$s = a - b \log_{10}(V_{\text{flat}})$	26	$s = 1.770 - 0.795 \log_{10}(V_{\text{flat}})$	0.588
κ-eligible subset	$s = a - b \log_{10}(V_{\text{flat}}) + d \log_{10}(\text{Rdisk})$	26	$s = 1.831 - 0.834 \log_{10}(V_{\text{flat}}) + 0.044 \log_{10}(\text{Rdisk})$	0.590

The baseline relation with Vflat alone is already strong. In the full accepted sample it explains about 54% of the variance in outer slope (R² = 0.544). In the κ-eligible subset the relation remains strong and in fact becomes slightly tighter (R² = 0.588). Adding log10(Rdisk) produces only a marginal improvement in either sample. This means that a future κ term will only be scientifically useful if it improves the fit beyond a relation that is already substantially controlled by Vflat.

5. Interpretation for PIUM

The present calculation does not yet validate the full PIUM relation $s = a - b \log_{10}(V_{\text{flat}}) - c\kappa$, because κ has not been measured. What it does establish is that the proposed PIUM framework has a viable empirical entry point. The scale-dependent term $-b \log_{10}(V_{\text{flat}})$ is already supported by the accepted SPARC outer-slope data, and this remains true after restricting the sample to galaxies whose geometry is suitable for future Fourier-based morphology analysis.

The practical consequence is clear. A κ program is worth pursuing, but only as a second-stage test. The immediate burden on κ is not to create the basic trend from nothing, but to determine whether morphological coherence provides measurable explanatory power beyond the already strong Vflat scaling. If κ improves the baseline model, then PIUM gains a concrete morphology–dynamics result. If κ does not improve the model, then the current PIUM morphology term should be reconsidered or redefined.

6. Rationale for exclusions

The exclusions should be interpreted methodologically rather than physically. An excluded galaxy is not 'irrelevant' to PIUM; it is only unsuitable for a first-pass κ estimate under conservative image-analysis assumptions.

Too small on the sky: Systems with Rdisk,app < 30 arcsec are likely to yield unstable low-order Fourier amplitudes because the coherent disk structure is sampled by too few independent resolution elements.

Edge-on geometry: For highly inclined systems, spiral and azimuthal modes are strongly projection-mixed. In such cases a measured Fourier pattern would reflect line-of-sight superposition at least as much as intrinsic morphology.

Face-on geometry: Nearly face-on systems suppress geometric leverage for deprojection, making the intrinsic orientation and true large-scale non-axisymmetric structure less constrained in a first-pass pipeline.

7. Appendix A. Galaxy-by-galaxy screening table

The following table records the first-pass κ-screening outcome for every galaxy in the accepted SPARC sample. All values are taken from the accepted outer-slope table except for the apparent scale length, which is computed from Rdisk and D.

Galaxy	Inc (deg)	D (Mpc)	Rdisk (kpc)	Rdisk app (")	Vflat (km/s)	Outer slope	κ screening
D631-7	59.0	7.72	0.70	18.7	57.7	0.483	too small on
DDO154	64.0	4.04	0.37	18.9	47.0	0.428	too small on
DDO161	70.0	7.50	1.22	33.6	66.3	0.375	Eligible for first-pass κ measurement
ESO116-G012	74.0	13.00	1.51	24.0	109.1	0.101	too small on
ESO563-G021	83.0	60.80	5.45	18.5	314.6	0.004	edge-on and too small
F563-1	25.0	48.90	3.52	14.8	109.9	0.173	face-on and too small
IC2574	75.0	3.91	2.78	146.7	66.4	0.614	Eligible for first-pass κ measurement
IC4202	90.0	100.40	4.78	9.8	242.6	-0.022	edge-on and too small
NGC0100	89.0	13.50	1.66	25.4	88.1	0.229	edge-on and too small
NGC0247	74.0	3.70	3.74	208.5	104.9	0.191	Eligible for first-pass κ measurement
NGC0289	46.0	20.80	6.74	66.8	163.0	0.010	Eligible for first-pass κ measurement
NGC0300	42.0	2.08	1.75	173.5	93.3	0.174	Eligible for first-pass κ measurement
NGC0891	90.0	9.91	2.55	53.1	216.1	-0.087	edge-on geometry
NGC1003	67.0	11.40	1.61	29.1	109.8	0.130	too small on
NGC1090	64.0	37.00	3.53	19.7	164.4	-0.059	too small on
NGC2403	63.0	3.16	1.39	90.7	131.2	0.102	Eligible for first-pass κ measurement
NGC2683	80.0	9.81	2.18	45.8	154.0	-0.208	edge-on geometry
NGC2841	76.0	14.10	3.64	53.2	284.8	-0.081	edge-on geometry
NGC2903	66.0	6.60	2.33	72.8	184.6	-0.113	Eligible for first-pass κ measurement
NGC2915	56.0	4.06	0.55	27.9	83.5	0.095	too small on
NGC3109	70.0	1.33	1.56	241.9	66.2	0.451	Eligible for first-pass κ measurement
NGC3198	73.0	13.80	3.14	46.9	150.1	-0.016	Eligible for first-pass κ measurement
NGC3521	75.0	7.70	2.40	64.3	213.7	-0.034	Eligible for first-pass κ measurement
NGC3726	53.0	18.00	3.40	39.0	168.0	0.035	Eligible for first-pass κ measurement
NGC3741	70.0	3.21	0.20	12.9	50.1	0.378	too small on
NGC3769	70.0	18.00	3.38	38.7	118.6	-0.037	Eligible for first-pass κ measurement
NGC3917	79.0	18.00	2.63	30.1	135.9	0.056	edge-on geometry
NGC3992	56.0	23.70	4.96	43.2	241.0	-0.117	Eligible for first-pass κ measurement
NGC4013	89.0	18.00	3.53	40.5	172.9	-0.058	edge-on geometry
NGC4088	69.0	18.00	2.58	29.6	171.7	-0.063	too small on
NGC4100	73.0	18.00	2.15	24.6	158.2	-0.167	too small on
NGC4157	82.0	18.00	2.32	26.6	184.7	-0.062	edge-on and too small
NGC4183	82.0	18.00	2.79	32.0	110.6	-0.001	edge-on geometry
NGC4214	15.0	2.87	0.51	36.7	80.1	0.076	face-on geometry
NGC4217	86.0	18.00	2.94	33.7	181.3	-0.111	edge-on geometry
NGC4559	67.0	9.00	2.10	48.1	121.2	0.092	Eligible for first-pass κ measurement
NGC5033	66.0	15.70	5.16	67.8	194.2	-0.149	Eligible for first-pass κ measurement
NGC5055	55.0	9.90	3.20	66.7	179.0	-0.110	Eligible for first-pass κ measurement
NGC5371	53.0	39.70	7.44	38.7	209.5	-0.207	Eligible for first-pass κ measurement
NGC5585	51.0	7.06	1.53	44.7	90.3	0.176	Eligible for first-pass κ measurement
NGC5907	88.0	17.30	5.34	63.7	215.0	-0.075	edge-on geometry
NGC5985	60.0	39.70	7.01	36.4	293.6	-0.002	Eligible for first-pass κ measurement
NGC6015	60.0	17.00	2.30	27.9	154.1	0.032	too small on
NGC6503	74.0	6.26	2.16	71.2	116.3	-0.005	Eligible for first-pass κ measurement
NGC6674	54.0	51.20	6.04	24.3	241.3	-0.087	too small on
NGC6946	38.0	5.52	2.44	91.2	158.9	-0.133	Eligible for first-pass κ measurement
NGC7331	75.0	14.70	5.02	70.4	239.0	-0.002	Eligible for first-pass κ measurement
NGC7793	47.0	3.61	1.21	69.1	0.0	-0.141	Eligible for first-pass κ measurement
NGC7814	90.0	14.40	2.54	36.4	218.9	-0.066	edge-on geometry
UGC00128	57.0	64.50	5.95	19.0	129.3	0.034	too small on
UGC02487	36.0	69.10	7.89	23.6	332.0	-0.116	too small on
UGC02885	64.0	80.60	11.40	29.2	289.5	0.095	too small on
UGC02916	50.0	65.40	6.15	19.4	182.7	-0.217	too small on
UGC02953	50.0	16.50	3.55	44.4	264.9	-0.117	Eligible for first-pass κ measurement
UGC03205	67.0	50.00	3.19	13.2	219.6	-0.040	too small on
UGC03546	55.0	28.70	3.79	27.2	196.9	-0.074	too small on
UGC03580	63.0	20.70	2.43	24.2	126.2	0.154	too small on
UGC05253	37.0	22.90	8.07	72.7	213.7	-0.191	Eligible for first-pass κ measurement
UGC05716	54.0	21.30	1.14	11.0	73.1	0.219	too small on
UGC05721	61.0	6.18	0.38	12.7	79.7	0.055	too small on
UGC05986	90.0	8.63	1.67	39.9	113.0	-0.066	edge-on geometry
UGC06446	51.0	12.00	1.49	25.6	82.2	0.101	too small on
UGC06614	36.0	88.70	5.10	11.9	199.8	0.124	too small on
UGC06786	64.0	29.30	3.60	25.3	219.4	-0.049	too small on
UGC06787	66.0	21.30	5.37	52.0	248.1	0.168	Eligible for first-pass κ measurement
UGC06983	49.0	18.00	3.21	36.8	109.0	-0.025	Eligible for first-pass κ measurement
UGC07125	90.0	19.80	3.38	35.2	65.2	0.124	edge-on geometry
UGC07603	78.0	4.70	0.53	23.3	61.6	0.169	edge-on and too small
UGC08286	90.0	6.50	1.05	33.3	82.4	0.123	edge-on geometry
UGC08490	50.0	4.65	0.67	29.7	78.6	0.021	too small on
UGC08550	90.0	6.70	0.45	13.9	56.9	0.159	edge-on and too small
UGC08699	73.0	39.30	3.09	16.2	182.4	-0.038	too small on
UGC09037	65.0	83.60	4.28	10.6	152.3	0.064	too small on
UGC09133	53.0	57.10	6.97	25.2	226.8	-0.114	too small on
UGC11455	90.0	78.60	5.93	15.6	269.4	-0.093	edge-on and too small
UGC11914	31.0	16.90	2.44	29.8	288.1	0.162	too small on
UGC12506	86.0	100.60	7.38	15.1	234.0	-0.111	edge-on and too small
UGC12732	39.0	13.20	1.98	30.9	0.0	0.273	Eligible for first-pass κ measurement
UGCA444	78.0	0.98	0.83	174.7	37.0	0.546	edge-on geometry

Sources. Accepted outer-slope sample: pium_sparc_accepted_galaxies.csv. Global galaxy metadata: SPARC Table1 (Lelli, McGaugh & Schombert 2016). Coordinate list for Spitzer SHA target retrieval: user-supplied NED-resolved positions.